Electricity
Problem Sheet 1 Answers

1 a 1 proton, 1 neutron in the nucleus and 1 electron in the electron shell
b 3 protons, 3 neutrons in the nucleus and 3 electrons in the electron shell
c 5 protons, 6 neutrons in the nucleus and 4 electrons in the electron shell
d 10 protons, 12 neutrons in the nucleus and 11 electrons in the electron shell

2 a is neutral (same number of protons as electrons)
b is neutral (same number of protons as electrons)
c positive (1 more proton than electrons)
d negative (1 more electron than protons)

3 a True
b False
c False
d False
e True
f False
g True
h True
I False

4 a The protons DO NOT MOVE. The electrons are transferred from the wool to the rod, thus producing a negative charge on the wool and a positive charge on the rod.
b

5 a silk and polythene have gained electrons.
b glass and styrene have lost negative charges.
c (i) glass and styrene have more protons than electrons
(ii) silk and polythene have more negative charges than positive charges.
d Glass would attract either silk or polythene.
e Charged polythene would repel silk.

6 a negative

b Electrons from perspex are transferred to the silk thus giving the silk a negative charge. The perspex is left with a negative charge because there is now a surplus of protons.

7 a The comb could have either a positive or negative charge on it.

b The charge on the comb would be positive.

8 Rod B is positive and rod A could be either negative or neutral.

9 a S could be either positive, negative or neutral.
T could be either positive, negative or neutral

b C could be either negative or positive
D could be either negative or positive

10 electrons
protons and neutrons
negative
positive positive
positively
negative
positive negative
negatively

11 Rod Y is negative, Rod Z is neutral

12

Rod	**A**	**B**	**C**	**D**
B	attracts			
C	repels	attracts		
D	repels	attracts	repels	
E	attracts	repels	attracts	attracts

13 a positive and negative
b negative charge
c repel
d attract
e negative
f attractive and repulsive
g repel charge

14 a A **compass** is used to determine the poles of a magnet.

b An **electroscope** is used to detect the charge on an object.

c Neutral objects have **no charge**.
d **Unlike** charges are found to attract one another.
e A **negative** object will repel a negative object.
f Protons can be found in the **nucleus** of an atom.
g When one object rubs another, **electrons** are often moved from one object to the other.
h If an object loses some of its electrons it would become **positively** charged.
i When a piece of matter has **less** protons than electrons in it, it is negative

15 a T
b F
c F
d F
e F
f T
g T
i F
j T
k T

16 protons, neutrons and electrons
positive negative no
electrons electric
conductor
insulator
conductors
glass, cloth, plastic, paper, wood, ceramic etc (non metals)